

DRY-RUN GUIDE | largest_number.py

Source: Python\2nd_Belt\largest_number.py

CORE BEHAVIOR

Scans values for the maximum; index variant also retains the position of that value.

REPRESENTATIVE INPUT

'[4,9,2,9]'.

STEP TRACE

Max changes from 4 to 9 at index 1; later values do not exceed it.

EXPECTED OUTPUT / RETURN

Maximum 9; first index 1 (check source for zero/one-based output).

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\2nd_Belt\largest_number.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Find the largest number in the list.
```

```
002 | # Add your solution here.
```

```
003 |
```

```
004 | def find_largest_index(arr):
```

```
005 |     if not arr:
```

```
006 |         return -1 # Return -1 if array is empty
```

```
007 |
```

```
008 |     max_index = 0
```

```
009 |     for i in range(1, len(arr)):
```

```
010 |         if arr[i] > arr[max_index]:
```

```
011 |             max_index = i
```

```
012 |
```

```
013 |     return max_index
```

```
014 |
```

```
015 | input_array = [10, 25, 7, 31, 18]
```

```
016 | output = find_largest_index(input_array)
```

```
017 | print("Output:", output)
```

```
018 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.